Xingtong Yu, Ph.D.

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Research Areas

Graph mining and learning, Prompt learning, Foundation model, AI4Finance.

Education

2019.09 – 2024.06 **Ph.D.**
School of Computer Science and Technology, University of Science and Technology of China.

2015.09 – 2019.07 **B.E.**
School of the Gifted Young, University of Science and Technology of China.

Experiences

2026.02 – present **Postdoctoral Researcher.**
Department of Systems Engineering and Engineering Management, The Chinese University of Hong Kong.

2024.09 – 2026.01 **Research Scientist.**
School of Computing and Information System, Singapore Management University.

2024.05 – 2024.08 **Visiting Research Student.**
Center for Spatial Information Science, The University of Tokyo.

2022.06 – 2024.03 **Visiting Research Student.**
School of Computing and Information System, Singapore Management University.

Research Papers

* *denotes co-first authors with equal contribution.*

(Co-)First Author Publications

1. **X. Yu**, C. Zhou, X. Zhang, and Y. Fang, "Molecular knowledge guided prompt adaptation for pre-trained 2d graph encoders," in *International Conference on Information and Knowledge Management (CIKM)*, 2026.
2. J. Zhang*, **X. Yu***, Y. Fang, R. Stouffs, and Z. Trivic, "Urbangraphembeddings: Learning and evaluating spatially grounded multimodal embeddings for urban science," in *the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)*, 2026, pp. 10 246–10 257.
3. **X. Yu**, Z. Gong, C. Zhou, F. Yuan, and Z. Xinming, "Samgpt: Text-free graph foundation model for multi-domain pre-training and cross-domain adaptation," in *the ACM Web Conference (WWW)*, 2025, pp. 1142–1153.
4. **X. Yu***, Z. Liu*, F. Yuan, and Z. Xinming, "Node-time conditional prompt learning in dynamic graphs," in *the International Conference on Learning Representations (ICLR)*, 20 pages, 2025.
5. **X. Yu***, J. Zhang*, Y. Fang, and R. Jiang, "Non-homophilic graph pre-training and prompt learning," in *the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)*, 2025, pp. 1844–1854.
6. **X. Yu**, C. Zhou, Z. Kuai, X. Zhang, and Y. Fang, "Gcot: Chain-of-thought prompt learning for graphs," in *the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)*, 2025, pp. 3669–3679.
7. **X. Yu**, Y. Fang, Z. Liu, and X. Zhang, "Hgprompt: Bridging homogeneous and heterogeneous graphs for few-shot prompt learning," in *the AAAI Conference on Artificial Intelligence (AAAI)*, 2024, pp. 16 578–16 586.
8. **X. Yu**, Z. Liu, Y. Fang, Z. Liu, S. Chen, and X. Zhang, "Generalized graph prompt: Toward a unification of pre-training and downstream tasks on graphs," in *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2024, pp. 6237–6250.

9. **X. Yu**, C. Zhou, Y. Fang, and X. Zhang, "Multigprompt for multi-task pre-training and prompting on graphs," in *the ACM Web Conference (WWW)*, 2024, pp. 515–526.
10. Z. Liu*, **X. Yu***, Y. Fang, and X. Zhang, "Graphprompt: Unifying pre-training and downstream tasks for graph neural networks," in *the ACM Web Conference (WWW)*, 2023, pp. 417–428.
11. **X. Yu***, Z. Liu*, Y. Fang, and X. Zhang, "Learning to count isomorphisms with graph neural networks," in *the AAAI Conference on Artificial Intelligence (AAAI)*, 2023, pp. 4845–4853.

Co-Author Publications

1. C. Du, Y. Liu, **X. Yu**, Z. Xu, Y. Liu, and T. Li, "A2GBD: Attack-agnostic graph backdoor defense," *in the ACM Web Conference (WWW)*, pp. 1229–1239, 2026.
2. Z. Wen, W. Wei, Y. Fang, **X. Yu**, H. Zhang, W. Zhu, and X. Zhang, "Implicit-knowledge visual question answering with structured reasoning traces," *Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5114–5124, 2026.
3. X. Xie, Z. Guo, B. Li, **X. Yu**, Z. Liao, Z. Kang, and Y. Fang, "Clustering as reasoning: A k-means interpretation of chain-of-thought graph learning," *International Conference on Machine Learning (ICML)*, pp. 139 428–139 444, 2026.
4. Y. Zhao, Z. Wang, **X. Yu**, X. Li, J. Liang, and R. Li, "Learnable game-theoretic policy optimization for data-centric self-explanation rationalization," *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, pp. 1159–1173, 2026.
5. Y. Fang, Y. Wu, **X. Yu**, and S. Pan, "Few-shot learning on graphs: From meta-learning to llm-empowered pre-training and beyond," *Companion Proceedings of the ACM on Web Conference (WWW) 2025*, pp. 9–12, 2025.
6. S. Tan, D. Li, R. Jiang, Z. Wang, **X. Yu**, and M. Okumura, "Taming recommendation bias with causal intervention on evolving personal popularity," *in the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)*, pp. 2790–2800, 2025.
7. P. Xia, **X. Yu**, M. Hu, L. Ju, Z. Wang, P. Duan, and Z. Ge, "Hgclip: Exploring vision-language models with graph representations for hierarchical understanding," *in the International Conference on Computational Linguistics (COLING)*, pp. 269–280, 2025.
8. Z. Zhuang, L. Liu, K. Guo, **X. Yu**, H. Qi, Y. Shen, and B. Yin, "Mdpm: Modulating domain-specific prompt memory for multi-domain traffic flow prediction with transformers," *Knowledge-Based Systems*, vol. 325, p. 113 881, 2025.
9. W. Zhang, X. Deng, B. Jia, **X. Yu**, Y. Chen, J. Ma, Q. Ding, and X. Zhang, "Pixel adapter: A graph-based post-processing approach for scene text image super-resolution," *in the ACM International Conference on Multimedia (MM)*, pp. 2168–2179, 2023.

Academic Services

Area Chair

2027 SIGKDD

Conference Program Committee Member

SIGKDD, AAAI
2026 SIGKDD, SIGIR, NEURIPS
2025 ICLR, ICML, SIGKDD, WWW, AAAI
2024 ICLR, ICML, NeurIPS, WWW, CIKM
2023 NeurIPS

Journal Reviewer

Information Fusion, TKDE